

Assignment_1_Introduction_to_R

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Welcome to the first assignment of Foundations of Data Science!

Instructions

- Write your code in the R code chunks provided. Make sure to comment your code.
- When you are finished with your assignment, upload this R Markdown file under the Assignments tab on our class canvas page.
- You can work with your classmates, but remember to write their name in the comments.
- If you are inspired by code you found online make sure you cite the website, comment your code, and make it your own. No copy and pasting.
- IMPORTANT: Explain what your code does in a sentence or two accompanying each answer. I want to know not just if your code works but if you understand WHY it works.

1) Below I have provided a list of random letters and numbers. Please conduct the following tasks on this list:

- a) What type of data is each of the “things” in the list? Write these types to a vector called “my_types”. Psssst here is a hint for how you might want to write this code: <https://www.geeksforgeeks.org/adding-elements-in-a-vector-in-r-programming-append-method/#>.
- b) Count how many of each data type appear in your vector.
- c) How many numbers in this list are greater than 5?
- d) Generate a factor of types from your “my_types” vector.
- e) Get the index (or indices) of the number “1” in the lst list.

```
lst <- list(4, 'j', 'Y', 5, 'f', 'K', 8, 'z', 'T', 'c', 0, 'B', 2, 'm',  
           'X', 9, 'd', 'V', 3, 'n', 'G', 6, 'h', 'J', 2, 'p', 'O', 1, 'q',  
           'R', 7, 's', 'D', 2, 'l', 'N', 0, 'w', 'E', 5, 'e', 'U', 4, 'i',  
           'P', 1, 'a', 'H', 8, 'y', 'S', 3, 'g', 'F', 9, 'b', 'M', 6, 'k', 'L', 7)
```

```
#lst <- list(4, 'j', "Y", 5, 1) ### Test list
```

```
# a) What type of data is each of the "things" in the list? Write  
# these types to a vector called "my_types".
```

```
my_types <- c()  
greater_than_five <- 0  
  
for (i in seq_along(lst)){  
  item_class <- class(lst[[i]])  
  my_types <- append(my_types, values = item_class)  
}
```

```
# b) Count how many of each data type appear in your vector.
```

```
counts <- table(my_types)
cat("Counts of number of types in vector 'my_types'", "\n")
```

Counts of number of types in vector 'my_types'

```
print(counts)
```

```
my_types
character    numeric
         40         21
```

c) How many numbers in this list are greater than 5?

```
for (i in seq_along(lst)){
  item_class <- class(lst[[i]])
  if (item_class == "numeric" & lst[[i]] > 5)
    greater_than_five <- greater_than_five + 1
}
```

```
cat("Number of 'numeric' type items in the list greater than five:", greater_than_five, "\n")
```

Number of 'numeric' type items in the list greater than five: 8

d) Generate a factor of types from your "my_types" vector.

```
my_types_factor <- as.factor(my_types)
```

```
cat("The number of levels in my 'my_types' vector are:", my_types_factor[1], "\n")
```

The number of levels in my 'my_types' vector are: 2

```
cat("The levels in my 'my_types' vector are:", levels(my_types_factor), "\n")
```

The levels in my 'my_types' vector are: character numeric

e) Get the index (or indices) of the number "1" in the lst list.

```
for (i in seq_along(lst)){
  if (lst[[i]] == 1)
    cat("The number 1 is at index:", i, "\n")
}
```

The number 1 is at index: 28

The number 1 is at index: 46

2) In the “data” folder of this repository there is a file called GSE165252_vst_PERFECT.txt. This is real RNA-seq data from a phase II clinical trial investigating the impact of combined impact PD-L1 inhibitor (atezolizumab) and neoadjuvant chemoradiotherapy on resectable esophageal adenocarcinoma patients. You can find a link to the data on GEO here: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE165252>. You can read about this study here: <https://aacrjournals.org/clincancerres/article/27/12/3351/671431/Neoadjuvant-Chemoradiotherapy-Combined-with>. Download the GSE165252_vst_PERFECT.txt file to your local machine.

- Read in this file. Make sure you take into account that the first row contains Ensembl IDs (gene names)!
- How many samples are in this data set?
- How many genes are in this data set?
- Make a histogram of the expression values of a gene of your choosing from this data set.

- e) Subset this data to contain only their first three samples and first 2000 genes. Write this subset to a file called “eso_sub.csv”.

```
rna_data <- read.table("GSE165252_vst_PERFECT.txt", header = TRUE)
number_of_rows <- nrow(rna_data)
cat("The number of genes in the 'rna_data':", number_of_rows, "\n")
```

The number of genes in the 'rna_data': 62300

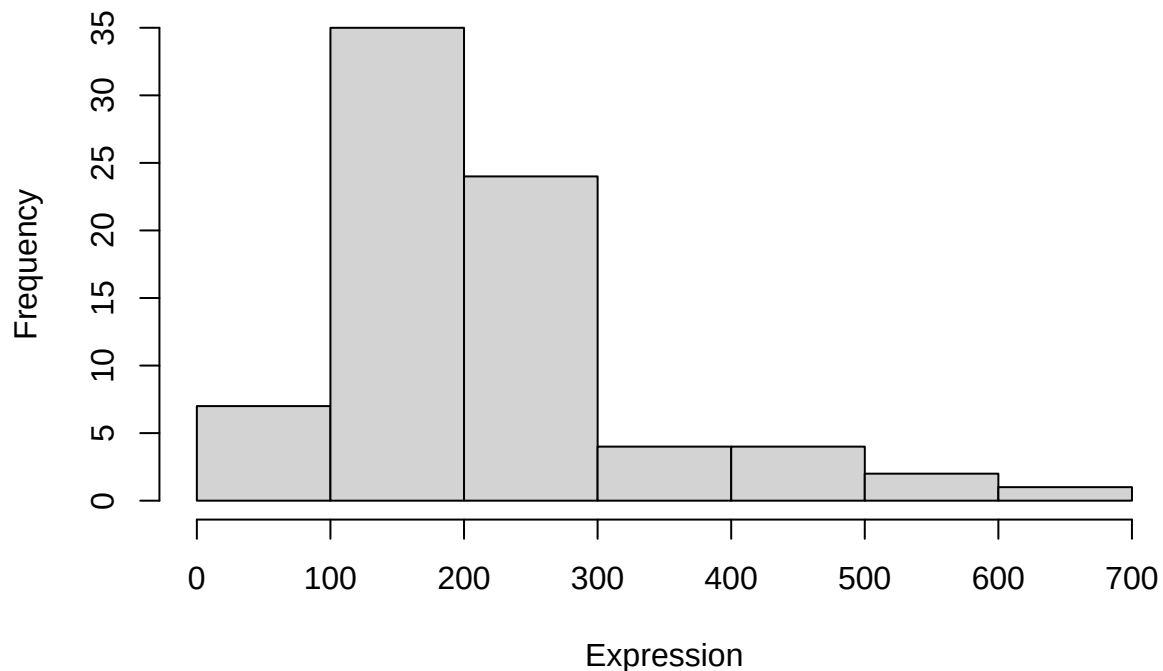
```
number_of_columns <- ncol(rna_data)
cat("The number of samples in the 'rna_data':", number_of_columns, "\n")
```

The number of samples in the 'rna_data': 77

```
#head(rna_data)
```

```
hist(as.numeric(rna_data[5,]),
     freq = TRUE,
     main = paste("Histogram of", rownames(rna_data)[5]),
     xlab = "Expression")
```

Histogram of ENSG00000000460.17



```
#eso_sub <- rna_data[1:2000, 1:3]
```

```
#write.csv(eso_sub, "eso_sub.csv", row.names = TRUE)
```

- 3) Write a function call “my_first_function”. The function should take the following vector: my_vector <- c(1,2,3,4,5,6,7,8,9). Your function should sort the numbers in my_vector into two categories: less than or equal to 5 or greater than 5. Hint: You can take a look at for loops and if/else statements to complete this question!

```

my_vector <- c(1,2,3,4,5,6,7,8,9)
ltoe5_vec <- c()
gt5_vec <- c()

my_first_function <- function(vec,x) {
  for (val in vec) {
    if (val <= x) {
      ltoe5_vec <- append(ltoe5_vec, values = val)
    }
    else{
      gt5_vec <- append(gt5_vec, values = val)
    }
  }
  return(list(ltoe_vector = ltoe5_vec, gtoe_vector = gt5_vec))
}

```

```

# Test the function
return_vectors <- my_first_function(my_vector,5)

print(return_vectors)

```

```

$ltoe_vector
[1] 1 2 3 4 5

```

```

$gtoe_vector
[1] 6 7 8 9

```